

VOLT-TERRA: County Charging Gap & Grid Feasibility Agentic Orchestrator

*A self-directed agent loop that ranks, scores, and justifies charging-gap priorities across counties via **Mireye API***

By Vedant Khanna | Mehtab Singh | Faaiz Hussain



The Problem

EV adoption and public charging buildout are both scaling fast, but they are not scaling together, and that mismatch is evident.

1. **Registrations are outrunning chargers in specific counties**, and nobody is tracking the ratio. State DMV data shows EV registrations per county growing year over year, and the DOE's Alternative Fuels Data Centre tracks charger locations, but the two **datasets are never joined**. A county can look fine in aggregate while a handful of underserved pockets go unnoticed until drivers are already circling for a plug.
2. A charger gap is not always a charger problem. Some underserved counties just need another station. Others sit far from any substation with spare capacity, and adding a charger there means a grid upgrade first, which is a different project, timeline, and budget entirely. **Funding bodies and utilities have no fast way to tell these two situations apart before sanctioning money.**

3. **Funding requests are hard to justify with a citable basis.** The reasoning is assembled by hand from scattered sources, which is slow and hard to defend under review.

Summary of the Solution

Sample prompt	Output
Run the charging-gap sweep for California and show me the top underserved county.	Madera County, CA- Grid Upgrade Priority Madera County shows 802 registered EVs per public charger, 3.3x the California state median, indicating a significant charging shortfall. However, the nearest substation (2.1 mi from county centroid, 115kV, PG&E territory) has only 0.3MW of spare interconnection capacity, insufficient to support standard DC fast-charging deployment without utility upgrade work. Recommendation: Prioritize this county for grid-capacity investment ahead of, or alongside, a hardware deployment, rather than a standalone NEVI/BEAD charger grant. Source: EIA Form 861 / CAISO Interconnection Queue, fetched 2026-08-14, confidence: high

VOLT-TERRA is an agentic evaluation layer built on top of the Mireye API that compresses weeks of manual research into a single, instant answer. It pinpoints exactly which counties have EV adoption outpacing their charging infrastructure and immediately separates the easy fixes from the heavy lifts.

Rather than guessing, teams can see at a glance which charging gaps just need a new station and which ones require a grid upgrade before anything else.

Key Signals

VOLT-TERRA identifies and sorts underserved counties in two steps:

Flag the gap: VOLT-TERRA tracks each county's driver-to-plug ratio (EV registrations per existing charger) over time. It flags a county once that ratio crosses a set multiple of the state median, keeping the threshold relative to regional peers rather than an arbitrary cutoff.

Sort the fix: Once a county is flagged as underserved, VOLT-TERRA uses Mireye grid data to assign it to one of two action categories:

Fund a charger now: Nearby substations have clear spare capacity, so the bottleneck is purely hardware and a standard charger deployment can move forward immediately.	Fund a grid upgrade first: Local grid infrastructure lacks the required capacity, meaning utility upgrades are needed before any hardware hits the ground.
---	---

By pairing relative demand thresholds with real-time grid availability, VOLT-TERRA routes funding straight to shovel-ready sites while flagging heavy-lift grid projects early.

Use Case Applications:

- **State Energy Offices:** Receive a prioritized, two-bucket county list complete with a one-paragraph justification memo per flagged county, ready to attach to a BEAD or NEVI funding request.
- **Utilities:** Gain early visibility into which service territories will require interconnection capacity planning ahead of demand.
- **Charging Network Operators:** Obtain a shortlist of counties where a new station is likely to see immediate utilization without waiting on a grid study.

Basically it's for the people who want to build EV charging stations but don't exactly know where they should build them and this solves the problem.

Additional Datasets

VOLT-TERRA pulls in **two external datasets that sit outside Mireye entirely: DOE's Alternative Fuels Data Center (AFDC) charger location data, and state DMV EV registration records**, 2–3 years back for the pilot state. Neither of these carries citation-grade provenance on its own, and they update on different schedules across

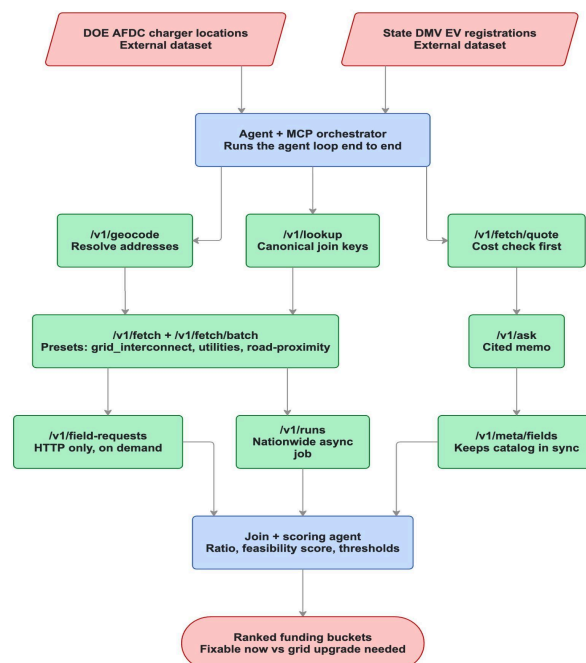
states, so the system timestamps them per county rather than treating them as current by default.

Inner workings of API calls

Once a county's centroid and corridor points are resolved, Mireye supplies the grid and infrastructure side of the picture: `/v1/geocode` and `/v1/lookup` turn addresses and coordinates into canonical join keys so the AFDC/DMV data lines up cleanly with Mireye's fields, and `/v1/fetch` (batched via `/v1/fetch/batch`, with `/v1/fetch/quote` checking cost before the sweep runs) pulls the `grid_interconnect` and `utilities` preset fields substation distance, voltage, interconnection queue capacity, service territory each one carrying a source, source URL, fetch timestamp, and confidence bucket traceable back to EIA and FERC-regulated filings. `/v1/ask` then turns the **ranked, bucketed county list into the plain-English, cited justification memo**.

Agentic Looping

The **agentic part is the loop that ties all of this together** without a human driving each call: **the orchestrator** decides which counties to sample, geocodes and joins the external data against Mireye's fields, computes the registrations-per-charger ratio and grid-feasibility score, applies the peer-relative thresholds to flag underserved counties, sorts each one into "fund a charger now" versus "fund a grid upgrade first," and generates its own memo end to end, on the dashboard.



Workflow

VOLT's pipeline runs as a loop, not a single call. Each stage below is an endpoint and what it is used for.

- **POST /v1/geocode** - resolves any address-level registration records into coordinates before sampling, and reports how the coordinate was derived so low-confidence matches can be filtered out.
- **POST /v1/lookup** - turns a county centroid or highway-corridor point into canonical join keys (coordinate, county/tract) so external EV registration and AFDC data line up cleanly with Mireye fields.
- **POST /v1/fetch** - pulls the grid_interconnect, utilities, and road-proximity preset fields for a sample point, called in a loop across each county's centroid and corridor points to keep a statewide sweep affordable; the field catalogue's published per-field pricing is checked against the sweep size before it runs.
- **POST /v1/ask** - turns the ranked, bucketed county list into the plain-English, cited justification memo that goes into the funding request.
- **GET /v1/meta/fields** - keeps VOLT's field list, and its cost-per-field estimate, in sync with Mireye's catalogue as grid and utility fields are added or revised.
- **POST /v1/field-requests** - requests a field not yet in the catalogue, such as a specific utility's interconnection queue status, if that ever becomes available.

Outside Mireye, the loop also pulls DOE AFDC charger locations and state DMV EV registration data (the external datasets), computes the registrations-per-charger ratio and the grid-feasibility score, applies the underserved/fixable-vs-constrained thresholds, and ranks counties into the two funding buckets.

Dashboard: Presenting the Agent's Decisions

VOLT-TERRA's agent loop runs end to end on its own geocoding, joining, scoring, and bucketing every county without a human driving it turn by turn. The dashboard's job is to surface what the agent already decided and let an analyst inspect and act on it, not to be a conversational front end for calling Mireye by hand.

What this looks like:

- **Ranked county view** - the two funding buckets (fund a charger now / fund a grid upgrade first) render as a sortable, filterable table, driven entirely by the agent's own scoring; nothing in this view is manually curated.
- **County drill-down** - clicking a flagged county surfaces the exact registrations-per-charger ratio, the grid_interconnect fields that decide its bucket, and the per-field citation source, source URL, fetched_at, confidence behind each number.
- **Memo panel** - the /v1/ask-generated justification memo for that county, ready to copy into a BEAD or NEVI funding request without leaving the dashboard.
- **Re-run trigger** - a button that re-runs the agent loop for a single county or the full state sweep on demand, instead of the analyst re-issuing API calls by hand.

The dashboard stays a thin, read-only layer over the agent's output: the agent decides, the dashboard displays the decision and lets a human act on it, attach the memo, request a re-run, and flag a county for manual review. None of the join logic, scoring, or thresholding runs in the dashboard itself; it's rendering what the backend agent loop already computed.

14-Day Development Plan

- **Days 1–2** - Pull and clean the two external datasets: DOE AFDC charger locations and state DMV EV registration data for the pilot state, 2–3 years back.
- **Days 3–4** - Auth and pilot setup: get Mireye API access working end-to-end, choose a pilot state and confirm its DMV data is usable at the county level.
- **Days 5–7** - Build the join pipeline: compute registrations-per-charger ratio per county, sample each county's centroid and corridor points, and pull grid_interconnect and utilities fields via [/v1/fetch](#), looped across samples..
- **Days 8–9** - Build the signal logic: peer-relative underserved threshold, grid-feasibility score, and the fixable-now versus grid-constrained split.
- **Days 10–11** - Wire up the dashboard: ranked county view, drill-down with per-field citations, and the per-county justification memo generated via [/v1/ask](#).
- **Days 12–13** - Backtest against 10–20 counties with known NEVI/BEAD outcomes, and handle edge cases such as counties with sparse charger data.
- **Day 14** - Build a product demo and ship.